

Blue/Green Deployments with Elastic Beanstalk

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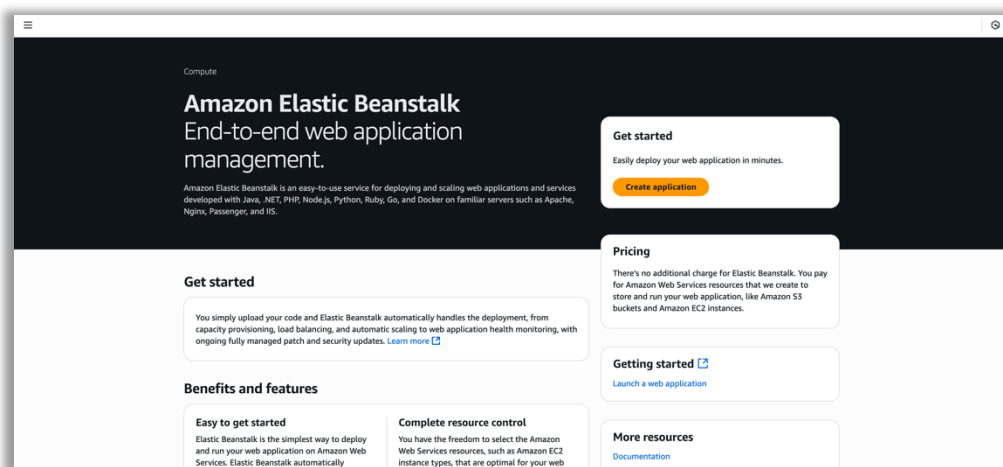
Introduction

This project demonstrates the implementation of a Blue/Green deployment strategy using AWS Elastic Beanstalk. The goal was to avoid disruptions to live applications while testing new changes. The process involved creating and cloning Elastic Beanstalk environments, uploading application versions, and performing a CNAME swap to transition traffic.

Steps Completed

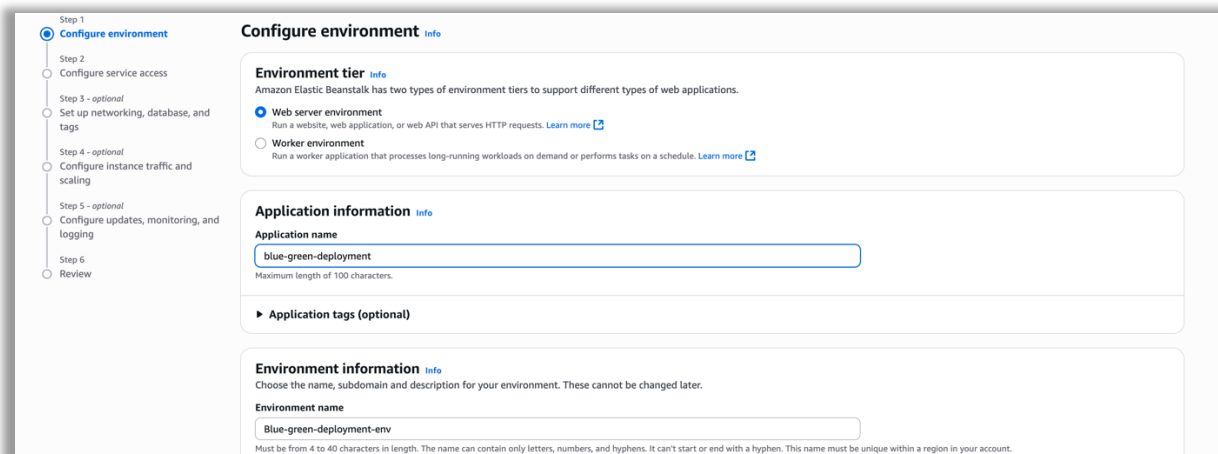
1. Log In to AWS Console

- Accessed the AWS Management Console using provided credentials.
- Ensured the selected region was us-east-1.
- Opened the Elastic Beanstalk service.



2. Create Elastic Beanstalk Environment

- Clicked Create application.
- Entered blue-green-deployment as the Application name.
- Chose a unique domain (e.g., blue1234) and verified availability.
- Set environment name to Blue-Deployment.



- Selected the Python platform.
- Uploaded the blue_app.zip file.
- Assigned the version label blueish.

Platform Info

Platform type

- ☒ Managed platform
Platforms published and maintained by Amazon Elastic Beanstalk. [Learn more](#)
- ☐ Custom platform
Platforms created and owned by you. This option is unavailable if you have no platforms.

Platform
Python

Platform branch
Python 3.13 running on 64bit Amazon Linux 2023

Platform version
4.5.2 (Recommended)

Application code Info

- ☐ Sample application
- ☐ Existing version
Application versions that you have uploaded.
- ☒ Upload your code
Upload a source bundle from your computer or copy one from Amazon S3.

Version label
Unique name for this version of your application code.
blueish

Source code origin. Maximum size 500 MB

- ☒ Local file
- ☐ Public S3 URL

Upload application

[Choose file](#)

File name: blue_app.zip
File must be less than 500MB max file size

- Chose Custom configuration.
- Selected the existing service role aws-elasticbeanstalk-ec2-role.
- Chose the custom VPC LabVPC, activated Public IP, and selected subnets.

Configure service access Info

Service access
IAM roles, assumed by Elastic Beanstalk as a service role, and EC2 instance profiles allow Elastic Beanstalk to create and manage your environment. Both the IAM role and instance profile must be attached to IAM managed policies that contain the required permissions. [Learn more](#)

Service role
Choose an IAM role for Elastic Beanstalk to assume as a service role. The IAM role must have the required IAM managed policies.

aws-elasticbeanstalk-service-role2 [Create role](#)

EC2 instance profile
Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.

aws-elasticbeanstalk-ec2-role [Create role](#)

EC2 key pair - optional
Select an EC2 key pair to securely log in to your EC2 instances. [Learn more](#)

Choose a key pair

Set up networking, database, and tags - optional Info

Virtual Private Cloud (VPC)

VPC
Launch your environment in a custom VPC instead of the default VPC. You can create a VPC and subnets in the VPC management console. [Learn more](#)

vpc-00646ed6c4decefa0 | (10.0.0.0/16) | LabVPC [Create custom VPC](#)

Instance settings
Choose a subnet in each AZ for the instances that run your application. To avoid exposing your instances to the Internet, run your instances in private subnets and load balancer in public subnets. To run your load balancer and instances in the same public subnets, assign public IP addresses to the instances. [Learn more](#)

Public IP address
Assign a public IP address to the Amazon EC2 instances in your environment.

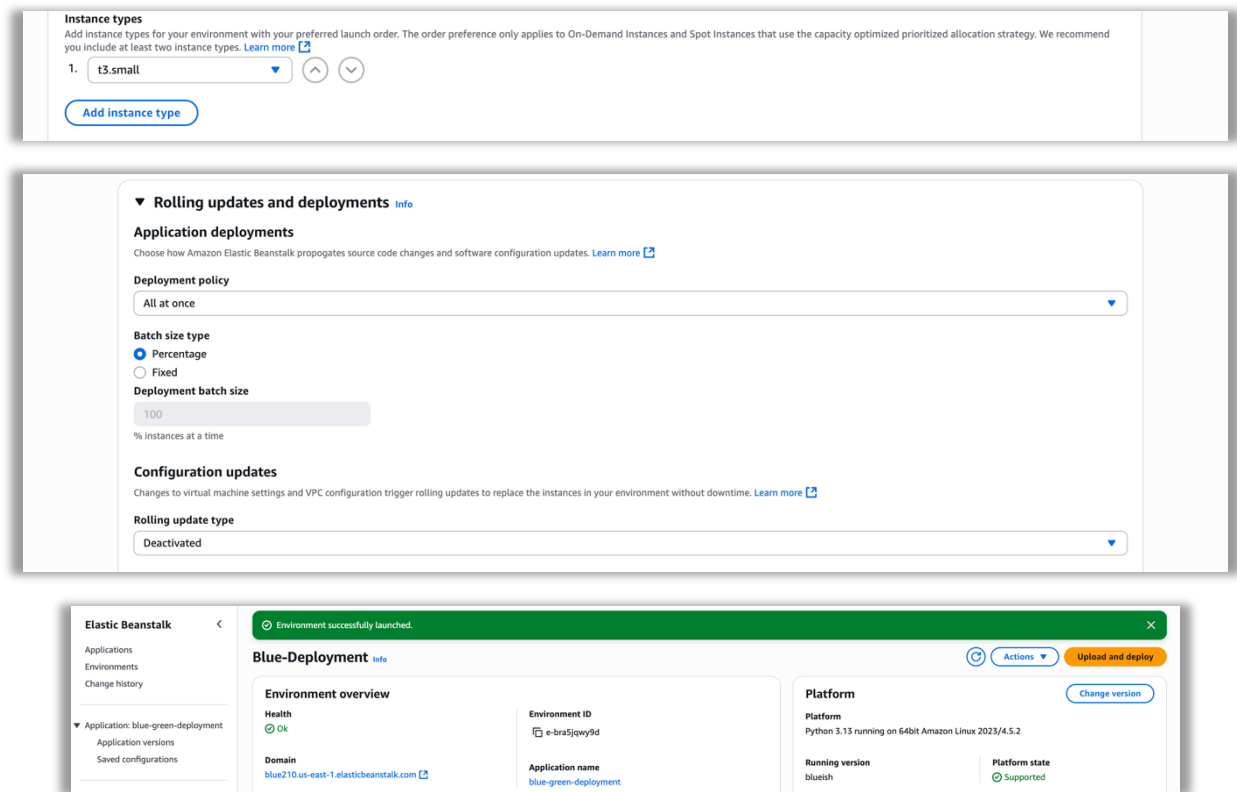
☒ Activated

Instance subnets

Filter instance subnets

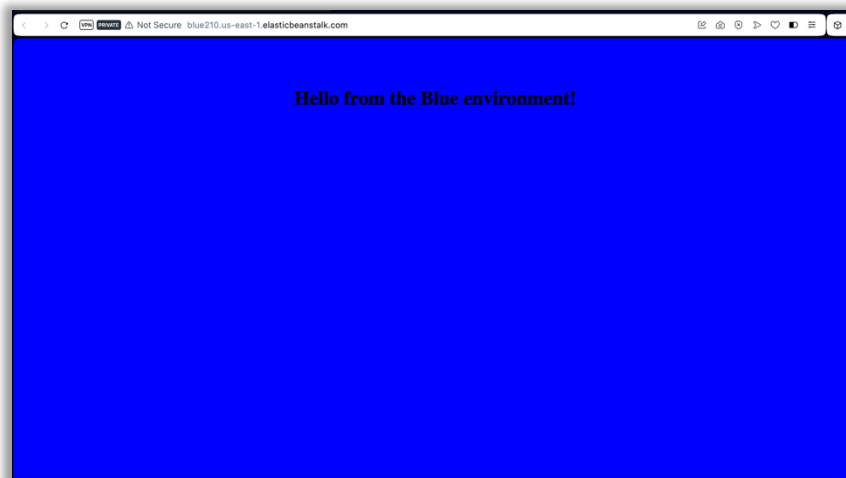
Availability Zone	Subnet	CIDR	Name
☒ us-east-1a	subnet-0ec95e6ae0bf295d	10.0.1.0/24	LabPublicSubnet

- Removed the t3.micro instance type.
- Skipped logging and monitoring configuration.
- Submitted the deployment and verified the application was correctly deployed via the domain link.



3. Clone Elastic Beanstalk Environment

- After confirming Blue-Deployment was active, selected Actions > Clone Environment.
- Renamed environment to Green-Deployment.
- Entered a unique domain (e.g., green5678) and verified availability.
- Updated the environment description.
- Selected the same service role (aws-elasticbeanstalk-ec2-role).
- Initiated the clone process.



Clone environment [Info](#)

You can launch a new environment based on an existing environment's configuration settings while optionally choosing a different platform version for the new environment. [Learn more](#)

Original environment

Environment name:
Blue-Deployment

Environment domain:
blue210.us-east-1.elasticbeanstalk.com

Platform:
Python 3.13 running on 64bit Amazon Linux 2023/4.5.2

New environment

Before cloning your environment, check your environment's stack drift status in CloudFormation.
Out of sync changes will not be carried over to the cloned environment. For critical resources, like ingress security groups, you may need to reestablish them in the cloned environment. [Learn more](#)

[Detect stack drift](#)

Environment name
Green-deployment-env

Must be from 4 to 40 characters in length. The name can contain only letters, numbers, and hyphens. It can't start or end with a hyphen. This name must be unique within a region in your account.

Elastic Beanstalk

Applications
Environments
Change history

▼ Application: blue-green-deployment
Application versions
Saved configurations

▼ Environment: Green-deployment-env

Environment successfully launched.

Green-deployment-env [Info](#)

Environment overview

Health
Pending

Domain
green210.us-east-1.elasticbeanstalk.com

Environment ID
e-rticvtnxg

Application name
blue-green-deployment

Platform

Platform
Python 3.13 running on 64bit Amazon Linux 2023/4.5.2

Running version
-

Platform state
Supported

[Actions](#) [Upload and deploy](#) [Change version](#)

4. Deploy Green Application

- Once Green-Deployment was active, clicked Upload and deploy.
- Uploaded the green_app.zip file.
- Labeled the version Green.
- Deployed the green application version.
- Verified deployment via the domain link.

Green-deployment-env [Info](#)

Environment overview

Health
Ok

Domain
green210.us-east-1.elasticbeanstalk.com

Events (11) [Info](#)

Filter events by text, property or value

Time

May 26, 2025 14:32:55 (UTC+1)

May 26, 2025 14:32:53 (UTC+1)

Upload and deploy

To deploy a previous version, go to the [Application versions page](#)

Upload application
[Choose file](#)

File must be less than 500MB max file size

Version label
Unique name for this version of your application code.
Version label

Current number of EC2 instances: 1

[Cancel](#) [Deploy](#)

Application available at green210.us-east-1.elasticbeanstalk.com.

Environment successfully launched.

Swap environment domain [Info](#)

When you swap an environment's domain with another environment's domain, you can deploy versions with no downtime. [Learn more](#)

Swapping the environment domain will modify the Route 53 DNS configuration, which may take a few minutes. Your application will continue to run while the changes are propagated.

Environment domain swap

Environment
Blue-Deployment (e-bra5jpy9d)

Environment domain:
blue210.us-east-1.elasticbeanstalk.com

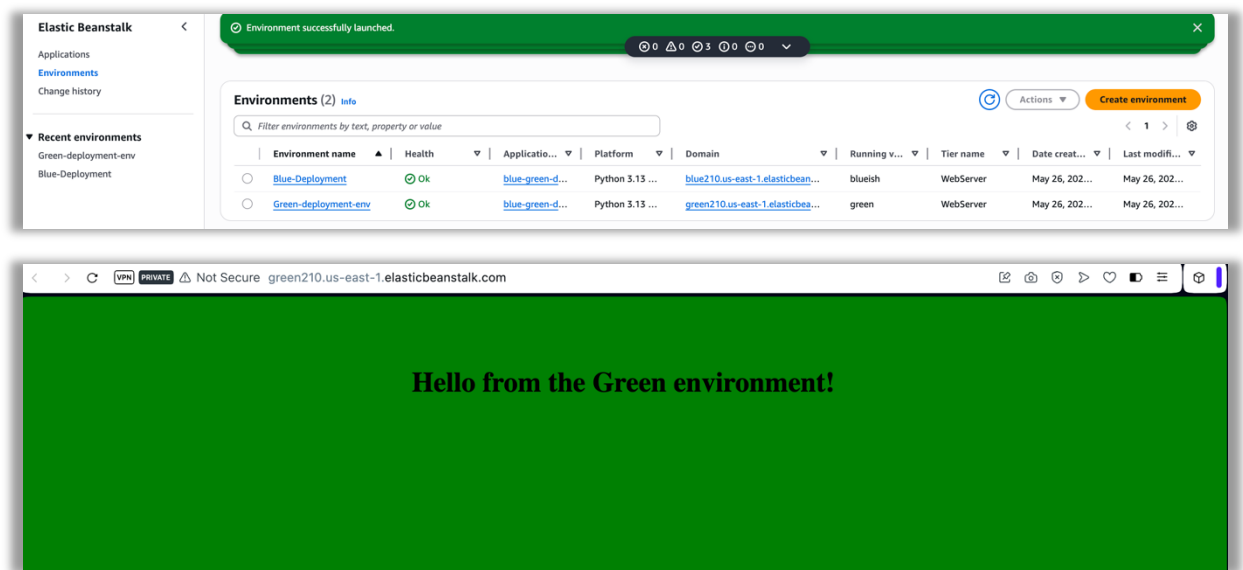
Environment domain to swap
Traffic will be redirected to the domain of the chosen environment.

Environment:
Green-deployment-env (e-rticvtnxg)

[View all environments](#)

Environment domain:
green210.us-east-1.elasticbeanstalk.com

[Cancel](#) [Swap](#)



5. Swap CNAMEs

- Navigated to the Elastic Beanstalk environments dashboard.
- Selected Blue-Deployment and clicked **Actions > Swap environment domain**.
- Chose Green-Deployment and confirmed the swap.
- Verified that the Blue-Deployment domain now pointed to the green application.

Conclusion

The blue/green deployment was successfully implemented using AWS Elastic Beanstalk. The environment was duplicated, modified, and tested independently before traffic was redirected using a CNAME swap. This approach ensured zero-downtime deployment and a rollback-safe update mechanism.

The screenshot shows the AWS Elastic Beanstalk console 'Events' page. The table has columns for Time, Type, and Details. Two events are listed:

Time	Type	Details
May 26, 2025 14:35:11 (UTC+1)	INFO	Updating environment Green-deployment-env's configuration settings.
May 26, 2025 14:35:03 (UTC+1)	INFO	Environment update is starting.